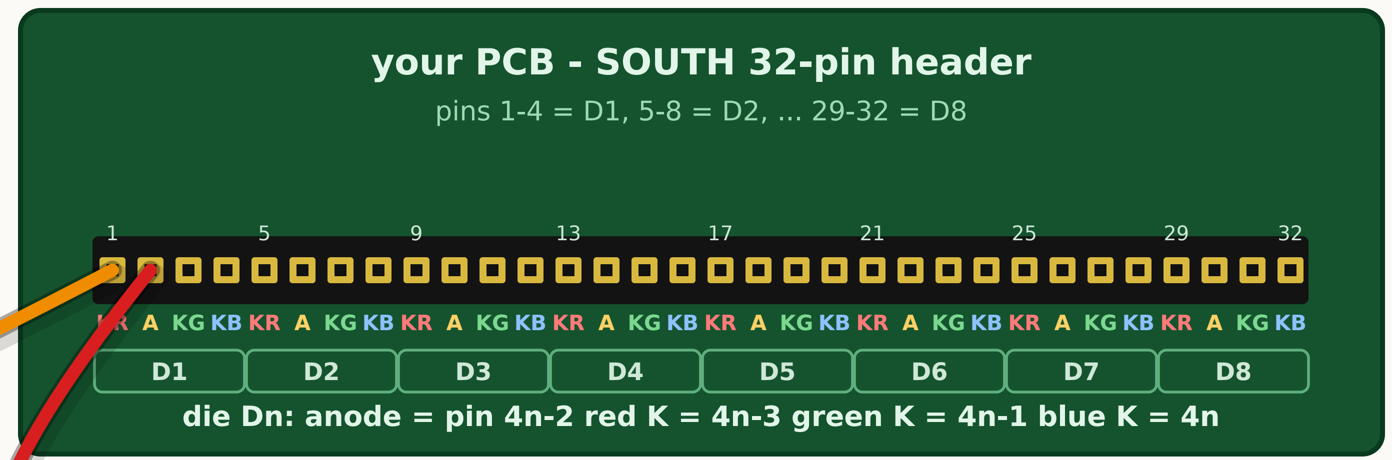
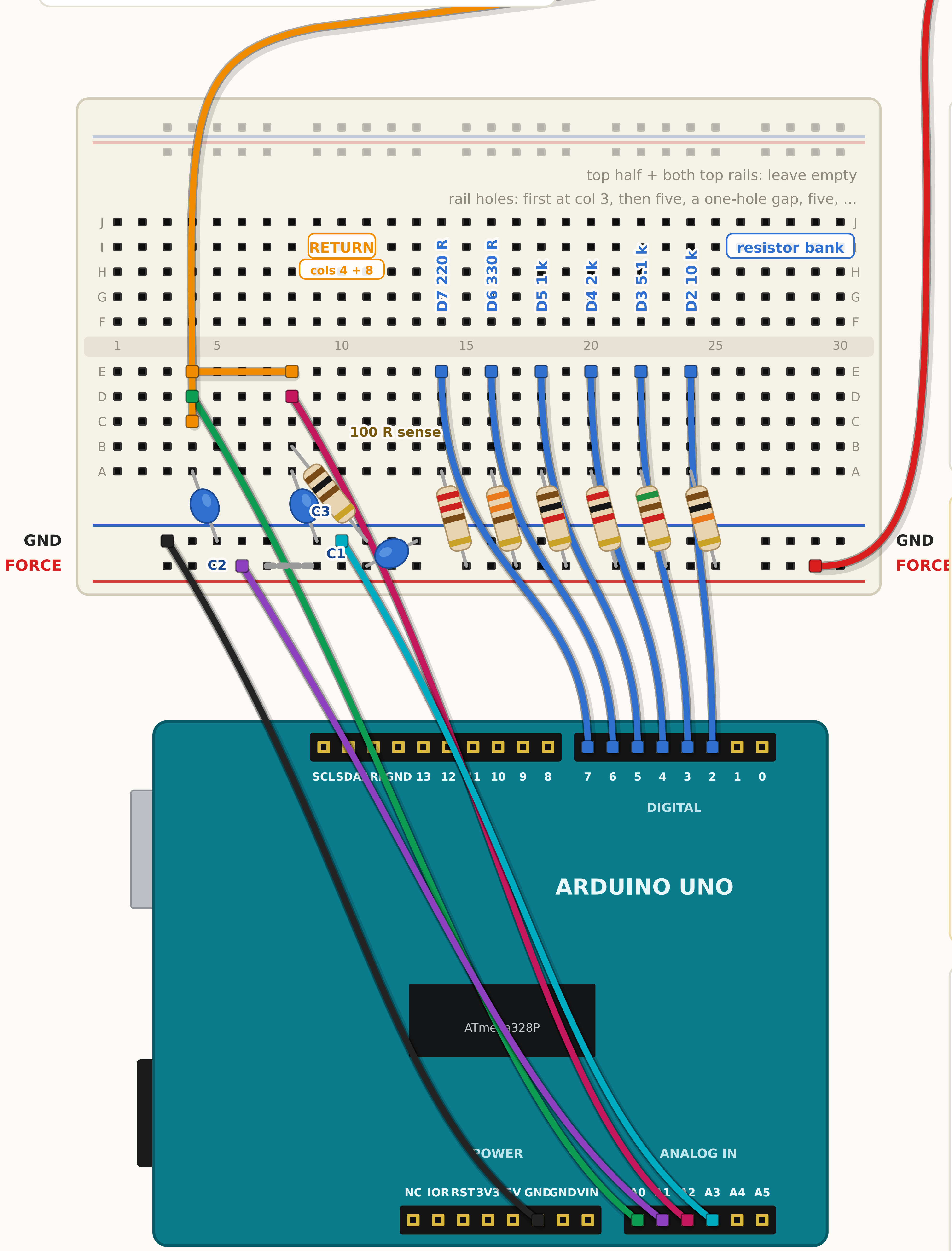
Arduino I-V rig - build this

Top view, drawn hole for hole. Only the bottom half of the breadboard is used.

build it in this order

1. six bank resistors: row A col 14..24, other leg FORCE one col right
2. jumper D2..D7 into row E of its own bank column
3. link E4 to E8 - all ten of those holes are now RETURN
4. 100 R from B8 down to the GND rail
5. C2 from A4, C3 from A8 to GND rail; C1 across the rails (68 nF)
6. jumper UNO GND to the GND rail col 3 - the only ground wire
7. jumper A0 to D4, A2 to D8, A1 to FORCE col 6, A3 to GND col 10
8. two F/M jumpers out to the die you are testing





every wire, by colour

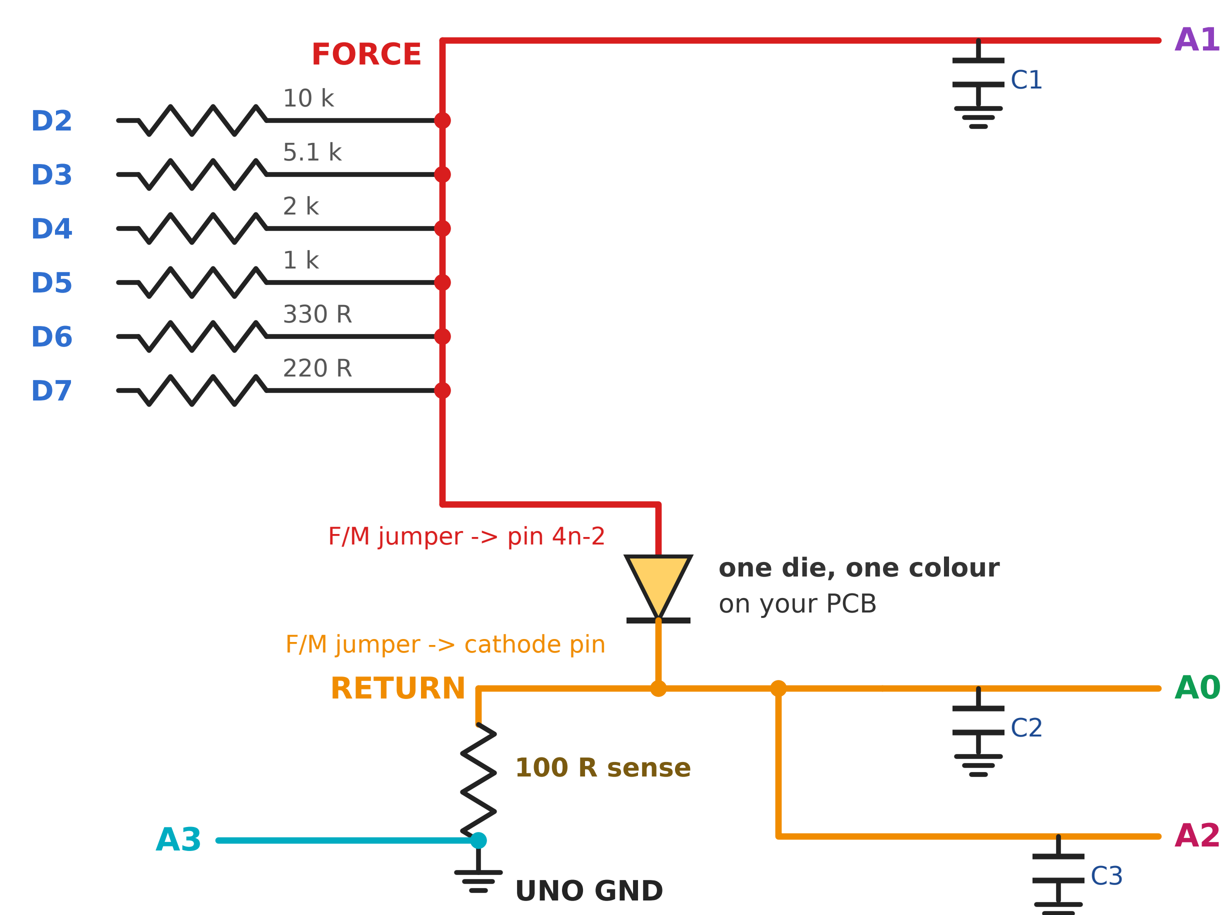
- 6 x UNO D2..D7 -> row E, cols 24 22 20 18 16 14
- 1 x UNO GND -> GND rail, col 3
- 1 x UNO A0 -> RETURN, hole D4
- 1 x UNO A1 -> FORCE rail, col 6
- 1 x UNO A2 -> RETURN, hole D8
- 1 x UNO A3 -> GND rail, col 10 NEW
- 1 x link E4 - E8 (RETURN becomes one node)
- F/M FORCE col 29 -> die anode, pin 4n-2
- F/M RETURN, C4 -> die cathode you are testing

check before you power up

- 12 male-male jumpers: 6 bank + A0 A1 A2 A3 + GND + 1 link
- 2 female-male jumpers - the only wires leaving the board
- 7 resistors: six in the bank, one 100 R RETURN to GND rail
- 3 caps, 68 nF each: C1 rail to rail, C2 + C3 on RETURN
- nothing in series between an ADC pin and the node it reads
- UNO GND touches the breadboard exactly once
- A0 and A2 land on the same node - that is intended
- A3 reads the GND rail: it cancels the ADC offset
- rail holes line up with the columns, starting at col 3
- each rail is one strip, so any hole on it is the same node
- the grey dash is only for boards with split rails

To change channel: move only the two female-male jumpers.

the same thing as a circuit



A0 and A2 sit on the same node, so they read the same voltage.

i = (A2 - A3) / R_SENSE. The difference cancels the ADC offset.